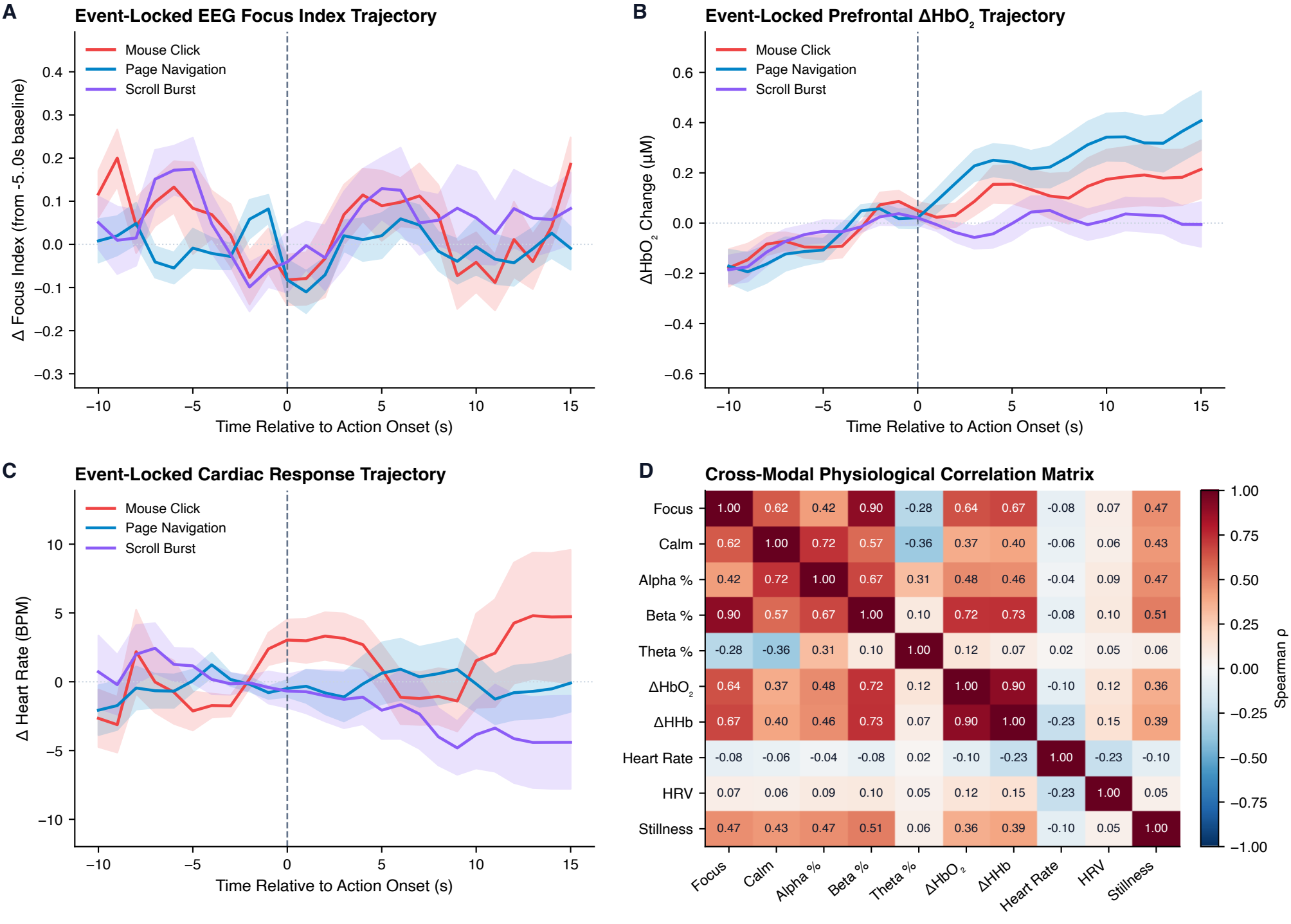
A
Event-Locked EEG Focus Index Trajectory
Mouse Click
Page Navigation
Scroll Burst
Δ Focus Index (from -5..0s baseline)
Time Relative to Action Onset (s)
B
Event-Locked Prefrontal ΔHbO2 Trajectory
Mouse Click
Page Navigation
Scroll Burst
ΔHbO2 Change (μM)
Time Relative to Action Onset (s)
C
Event-Locked Cardiac Response Trajectory
Mouse Click
Page Navigation
Scroll Burst
Δ Heart Rate (BPM)
Time Relative to Action Onset (s)
D
Cross-Modal Physiological Correlation Matrix
Focus 1.00 0.62 0.42 0.90 -0.28 0.64 0.67 -0.08 0.07 0.47
Calm 0.62 1.00 0.72 0.57 -0.36 0.37 0.40 -0.06 0.06 0.43
Alpha % 0.42 0.72 1.00 0.67 0.31 0.48 0.46 -0.04 0.09 0.47
Beta % 0.90 0.57 0.67 1.00 0.10 0.72 0.73 -0.08 0.10 0.51
Theta % -0.28 -0.36 0.31 0.10 1.00 0.12 0.07 0.02 0.05 0.06
ΔHbO2 0.64 0.37 0.48 0.72 0.12 1.00 0.90 -0.10 0.12 0.36
ΔHHb 0.67 0.40 0.46 0.73 0.07 0.90 1.00 -0.23 0.15 0.39
Heart Rate -0.08 -0.06 -0.04 -0.08 0.02 -0.10 -0.23 1.00 -0.23 -0.10
HRV 0.07 0.06 0.09 0.10 0.05 0.12 0.15 -0.23 1.00 0.05
Stillness 0.47 0.43 0.47 0.51 0.06 0.36 0.39 -0.10 0.05 1.00
Focus
Calm
Alpha %
Beta %
Theta %
ΔHbO2
ΔHHb
Heart Rate
HRV
Stillness
Spearman ρ